

Question-8.4.25

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Question

The equation of directrix of parabola $y^2 + 4y + 4x + 2 = 0$

- ① $x = -1$
- ② $x = \frac{-3}{2}$
- ③ $x = 1$
- ④ $x = \frac{3}{2}$

Solution

Given the parabola equation:

$$y^2 + 4y + 4x + 2 = 0 \quad (1)$$

The equation of a conic with directrix $\mathbf{n}^T \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0, \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, f = 2 \quad (2)$$

Find vertex $\mathbf{c}$ from

$$\mathbf{V} \mathbf{c} = -\mathbf{u} \quad (3)$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c_x \\ c_y \end{pmatrix} = - \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad (4)$$

$$\begin{pmatrix} 0 \\ c_y \end{pmatrix} = - \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad (5)$$

$$c_y = -2 \quad (6)$$

Solution

Put $c_y = -2$ into parabola to find c_x :

$$(-2)^2 + 4(-2) + 4c_x + 2 = 0 \Rightarrow c_x = \frac{1}{2}\mathbf{c} = \begin{pmatrix} \frac{1}{2} \\ -2 \end{pmatrix} \quad (7)$$

Eigenvector for zero eigenvalue of $\mathbf{V}$:

$$\mathbf{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} p = \frac{\|\mathbf{u}\|^2 - f}{4\mathbf{u}^T \mathbf{a}} = \frac{4 + 4 - 2}{8} = \frac{6}{8} = \frac{3}{4} \quad (8)$$

Equation of directrix (line perpendicular to $\mathbf{a}$, distance p from vertex):

$$\mathbf{n}^T \mathbf{x} = \mathbf{n}^T \mathbf{c} + p, \text{ where } \mathbf{n} = \mathbf{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mathbf{x} = \frac{1}{2} + \frac{3}{4} = \frac{5}{4} \quad (9)$$

Therefore, the equation of the directrix is approximately: $\mathbf{x} = \frac{3}{2}$

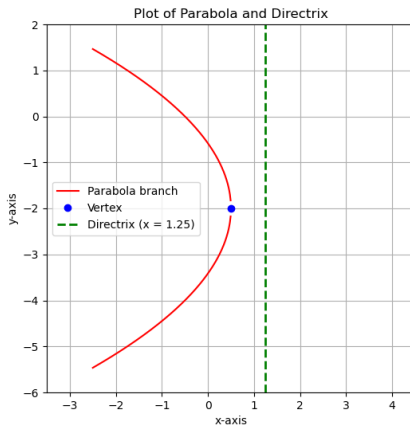


Figure:

```
// code.c
#include <stdio.h>

void get_parabola_data(double *vertex, double *p, double *
    directrix_x) {
    // vertex (1/2, -2)
    vertex[0] = 0.5;
    vertex[1] = -2.0;

    // parameter p = 3/4
    *p = 0.75;

    // directrix x = vertex_x + p
    *directrix_x = vertex[0] + *p;
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
import ctypes  
import numpy as np  
  
lib = ctypes.CDLL('./code.so')
```

```
get_parabola_data = lib.get_parabola_data
get_parabola_data.argtypes = [
    np.ctypeslib.ndpointer(dtype=np.float64, shape=(2,)),
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double)
]
get_parabola_data.restype = None

def get_data():
    vertex = np.zeros(2, dtype=np.float64)
    p = ctypes.c_double()
    directrix_x = ctypes.c_double()
    get_parabola_data(vertex, ctypes.byref(p), ctypes.byref(
        directrix_x))
    return vertex, p.value, directrix_x.value
```

```
import numpy as np
import matplotlib.pyplot as plt
from call import get_data

vertex, p, directrix_x = get_data()

# Generate parabola points from equation transformed for y in
  terms of x
# From  $(y+2)^2 = -4(x - 0.5)$ 
x = np.linspace(vertex[0] - 3, vertex[0] + 3, 400)
y_right = np.sqrt(-4 * (x - vertex[0])) - 2
y_left = -np.sqrt(-4 * (x - vertex[0])) - 2

fig, ax = plt.subplots(figsize=(8, 6))

# Plot parabola branches
ax.plot(x, y_right, 'r', label='Parabola branch')
ax.plot(x, y_left, 'r')
```

```
# Plot vertex
ax.plot(vertex[0], vertex[1], 'bo', label='Vertex')

# Plot directrix line: vertical line at x = directrix_x
ax.axvline(directrix_x, color='g', linestyle='--', linewidth=2,
           label=f'Directrix (x = {directrix_x:.2f})')

ax.set_xlim(vertex[0] - 4, vertex[0] + 4)
ax.set_ylim(vertex[1] - 4, vertex[1] + 4)
ax.set_aspect('equal')
ax.grid(True)
ax.legend()
ax.set_title('Plot of Parabola and Directrix')
ax.set_xlabel('x-axis')
ax.set_ylabel('y-axis')
plt.show()
```